

Question 1. (13 Points, 3/4/3/3 points per part)

Consider a neural network for a regression problem with response restricted in $(0,1)$. The network architecture is:

- Input dimension: $p=20$;
- Hidden layer 1: 15 units, ReLU activation, with parameters W_1, B_1 ;
- Hidden layer 2: 10 units, ReLU activation, with W_2, B_2 ;
- Output layer: 1 unit with sigmoid activation, with W_3, B_3 .

Answer the following questions.

- (a) Denote the input vector as x , Write down the mathematical form of the fitted response in terms of the network parameters.

- (b) Compute the total number of trainable parameters in this network.

- (c) This NN applies ridge regularization. State one advantage of ridge regularization. Denoting the observations and their fitted response as (x_i, y_i) and $\hat{y}_i$ respectively, write down the loss function L for this NN.

- (d) What modification can be done on this NN in order to change it into a logistic regression model? Then compare the original NN and the logistic regression, and state one advantage also one disadvantage of NN.

Modification:

One advantage of NN compared to the logistic regression:

One disadvantage of NN compared to the logistic regression:

Question 2. (12 Points ,8/4 Points Per Part)

A CNN is applied in a classification task. In the task, an 28×28 pixels RGB image is given, and the model is used to classify it among 3 possible classifiers. The structure of the CNN is described below:

- Input layer I
- Convolution layer C_1 : filter size 5×5 , 8 filters, stride=1, padding=0, activation ReLU;
- Maxpooling layer M : 2×2 , stride =2;
- Convolution layer C_2 : filter size 3×3 , 16 filters, stride=2, padding =1, activation ReLU;
- Global average pooling G : taking mean over the entire spatial channel;
- Output layer O , with softmax.

Answer the following questions:

- (a) Describe the dimension for each layer. Then find how many parameters are estimated in the model?

Dimension for each layer:

Layer I : $_\times_\times_\$ Layer C_1 : $_\times_\times_\$ Layer M : $_\times_\times_\$
Layer C_2 : $_\times_\times_\$ Layer G : $_\times_\times_\$ Layer O : $_\times_\times_\$

Total number of parameters for this CNN:

- (b) For one observation in classifier 1, in layer O , the classifier logits is $z = (1.2, 0.3, -0.5)$ for classes 1,2,3. Find the softmax probability for each classifier. Then find the loss for the fitting of this observation.

Question 3. (8 Points)

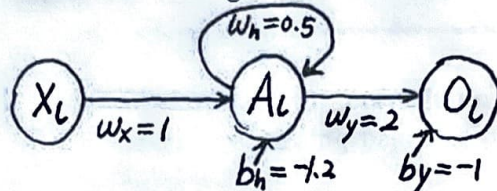
Consider the grayscale image of size 3×3 with input matrix X . A CNN is applied to the image. Its 1st convolution layer L_1 has stride 2, padding 1, 2 filters (W and W^T) and activation ReLU. The bias in L_1 are all 1. L_1 is followed by a maxpooling layer M : 2×2 , stride 2. X, W are given below. Find out the resulting units of M .

$$X = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{pmatrix}$$

$$W = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{pmatrix}$$

Question 4. (12 Points, 4/8 points per part)

A RNN is shown below. In the hidden layer, the activation function is $h_1(a) = |a|$ and the output layer uses activation sigmoid.



(a) Given data $x_1 = 1, x_2 = 2, x_3 = 1$, use them to predict $\hat{x}_4$

(b) Given that x_4 is observed to be 1, perform the backpropagation algorithm with learning rate 0.1 once, on the loss of predicting x_4 to update w_x .